From: **Austin Meyer** austin@x-plane.com
Subject: Re: blade radius 457 mm!
Date: September 4, 2026 at 12:45 PM
To: Jon Farhat j@jfi.net

Hmm ok let's see:

Here is my data on the latest prop, shown below.

Looks like:

```
tip_radius = 457.0, so the tip is 457 mm from the hub centerline
CENTER_HUB_to_FIBER = 31.0, so it is 31 mm from the hub out to the
first bit of carbon fiber (base of the grip)
so the blade itself SHOULD be 457 - 31 = 426 mm long
```

but you see 495 mm.. yikes!
that is clearly one of the earlier blades where i was pondering much longer radii, which I have indeed built!
but for speed and simplicity I am staying with the flite-proven 457's for your nacelles, just like I have with the align chassis, so we have the same power, rpm, rotor size, etc etc etc for rapid testing

so what I would EXPECT is 426 mm long blade... with 25 mm of sep between the inner and out mount points:

```
        const xdob CENTER_HUB_to_DIVOT1 =  40.0;
        const xdob CENTER_HUB_to_DIVOT2 =  65.0;
```

Here is that geometry with those specs... does this give you the dims we want?

Archive.zip

```
#if airplane_Ruby_vers_49
        const xdob cf_thick_per_side_R  =
one_layer_toray_t300_1K_plain   // 100% span

+ one_layer_uni                                 // 100% span

+ one_layer_toray_t300_1K_plain //  50% span

+ one_layer_uni                                 //  50% span

+ one_layer_toray_t300_1K_plain;//  25% span: i have had centrifugal
failure with the bearing pulling right through the CF, a perfect line
of pull thru

        const xdob cf_thick_per_side_T  = one_layer_toray_t300_1K_plain

+ one_layer_uni;

        const string version                    ="SOLID CORE V49-";
        const string pathR                              ="-TECH-CPP-
/Ruby-STL-GCODE/foils/GOE 601.txt";
        const string pathT                              ="-TECH-CPP-
/Ruby-STL-GCODE/foils/DEFIANT CANARD.txt";
        const xdob ref_chord                    =
35.0    ;               // ref chord in mm
```

```
        const xdob tip_radius                    =
457.0   ;               // tip radius from hub centerline in mm
        const xdob
RPM                                     =1800.0 ;               // rpm
        const xdob KTAS                                 =
35.0    ;               // KTAS

        const xdob AOA_root                             =
2.5;            // deg blade pitch at zero-pitch handle, for wash-out
        const xdob AOA_tip                              =
0.0;            // deg blade pitch at zero-pitch handle, for wash-out

        const xdob CENTER_HUB_to_FIBER  =  31.0;
        const xdob CENTER_HUB_to_DIVOT1 =  40.0;
        const xdob CENTER_HUB_to_DIVOT2 =  65.0;
        const xdob CENTER_HUB_to_GRIP0  =  80.0;

        const xdob bearing_OD1                  =
8.0;            // 5 mm bolt OD 8 mm
https://www.mcmaster.com/96977A212/
        const xdob bearing_OD2                  =
8.0;            // 5 mm bolt OD 8 mm
https://www.mcmaster.com/96977A212/

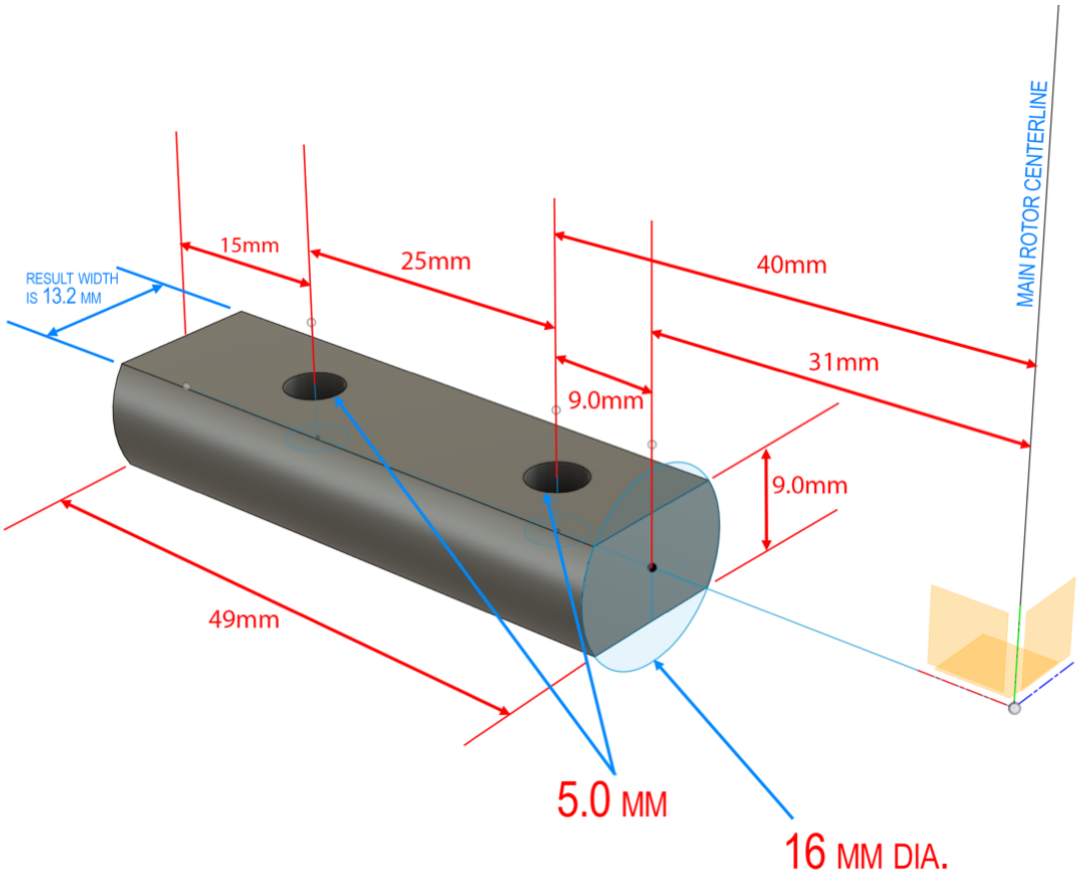
        const xdob gripper_hgt                  =   9.0 - 0.25; // i
built this to 9 mm and had to sand off some to make it fit! so make
this narrower moving forwards!

        const xdob metal_bushing_len     =  12.9;
        const xdob mold_depth_bot               =  (gripper_hgt/2) +
metal_bushing_len;      // table plane is at y = -mold_depth, so the
BOT mold table sits BELOW the blade at -17.275. we now SEE it as bot
too.
        const xdob mold_depth_top               =
-20.0;                                                          //
table plane is at y = -mold_depth, so the TOP mold table sits ABOVE the
blade at +20.0.  we now SEE it as top too.

#endif  airplane_Ruby_vers_49
```

Try it in X-Plane first.

> On Sep 3, 2026, at 7:58 PM, Jon Farhat <j@jfi.net> wrote:
>
> Regarding this… 457 blade radius. I'm a bit confused.
>
> If I read the last blade STL you gave me, (attached) it's 495mm x 18.36mm x 42.6mm.
>
> Add the 495 to the rotor head distance of 31mm to butt of blade, and you have a rotor radius of 562mm.
>
> If it's a 'blade radius' of 457, minus the 31, it's a rotor disc radius of only 426.
>
> Is the blade size (not blade radius) 457? If so, then add 31 to that and the true rotor disc radius is 488mm.



> On Aug 26, 2026, at 1:20 PM, Austin Meyer <austin@x-plane.com> wrote:
>
> blade radius 457 mm!
>
> un-changed!
>
> this WILL INCREASE LATER for sure!
>
> but we start with this to prove the concept!
>
> one it works, i ask you for ever-deep GEAR RATIOS as we go to ever-larger props!
>
> austin

Everyone is talking about socialism, but I don't know what socialism IS.